IP25-04: Automated Gleason Pattern Segmentation in Prostate Whole Slide Digitized Pathology Using Foundation Model Feature Embeddings

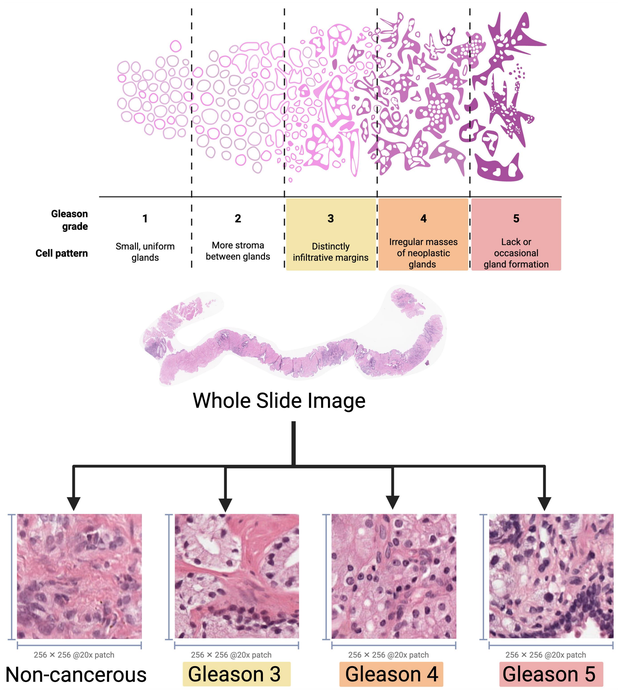
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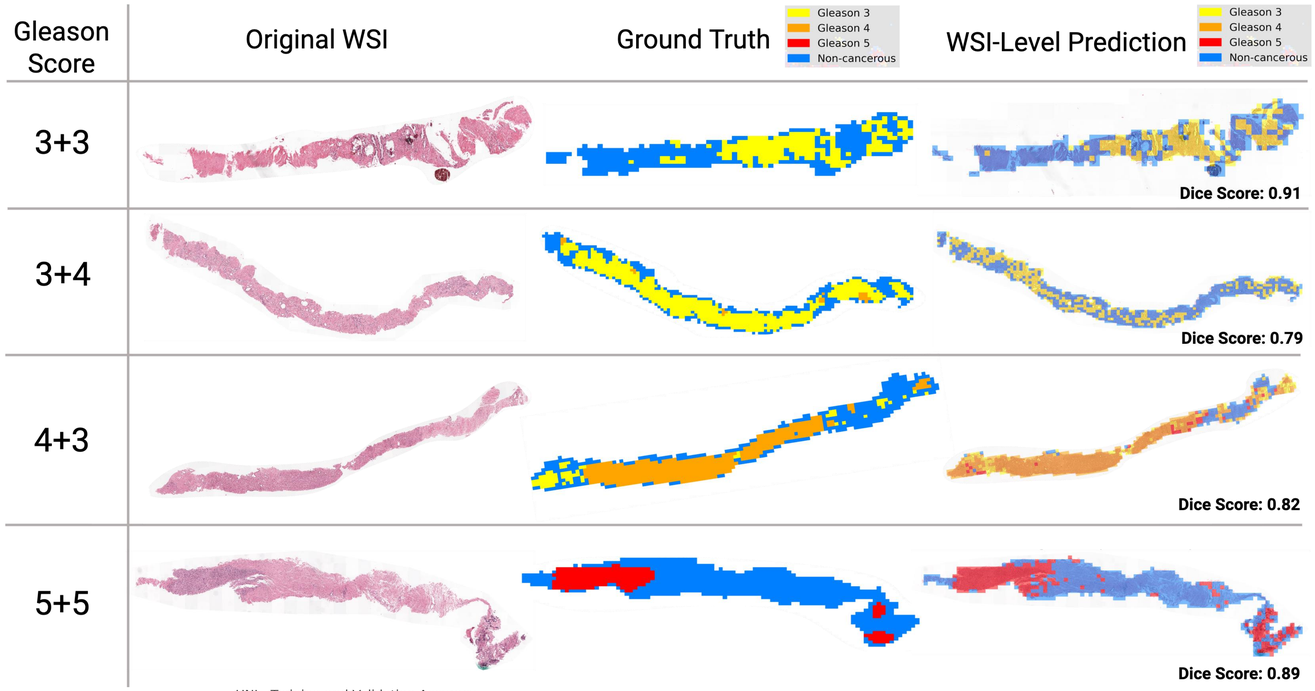
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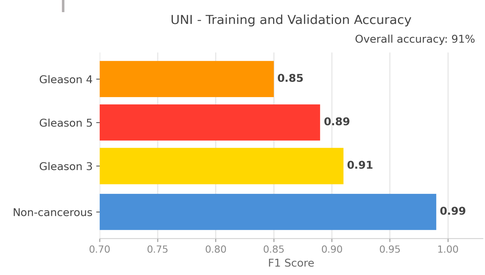
INTRODUCTION



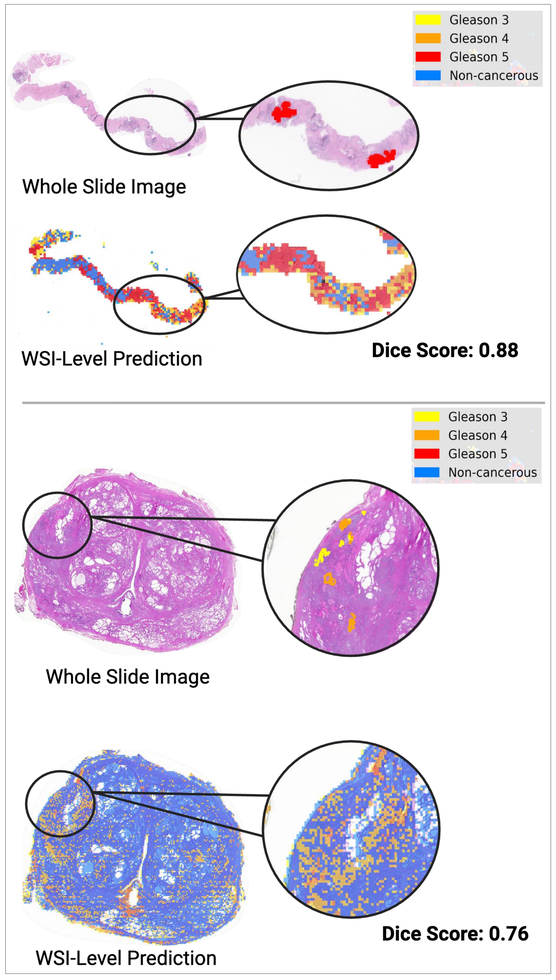
- Gleason grading of prostate cancer on H&E-stained whole slide images (WSIs) is standard of care but is time-consuming and subject to inter-observer variability
- Deep learning tools have shown promise for automated Gleason pattern segmentation, but require large amounts of task-specific labeled training data
- Foundation models (FMs), pretrained on large-scale datasets, can be fine-tuned for downstream tasks with limited supervision
- This study leverages **UNI**, a pathology-specific FM, for automated segmentation of Gleason patterns on digitized prostate needle core biopsy WSIs

RESULTS





- Prediction model (UNI) achieved **91% overall accuracy** in Gleason grade classification across 800 held-out validation patches
- Near-perfect identification of non-cancerous tissue (F1 = 0.99), minimizing false positive cancer calls
- Strong performance across all cancer grades
- WSI-level overlays demonstrate spatially coherent grade mapping across entire biopsy cores

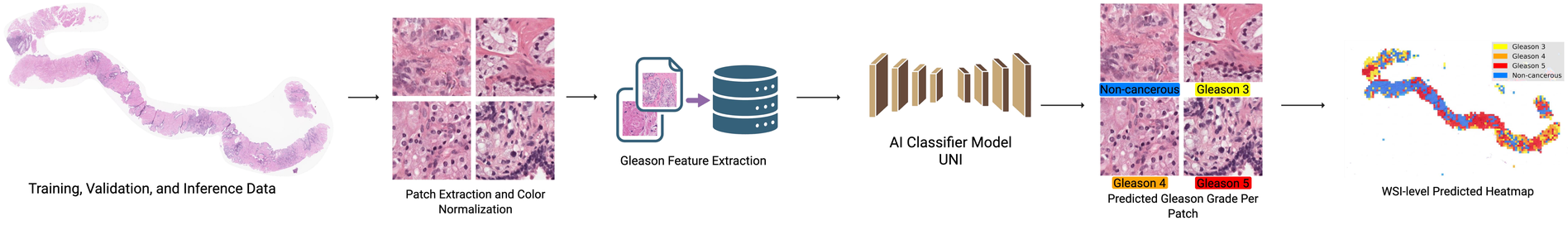


CONCLUSIONS

- A foundation model fine-tuned on prostate pathology achieves clinically relevant Gleason grading accuracy with limited labeled data Near-perfect non-cancerous detection reduces the risk of over-grading benign tissue
- Gleason 4 remains the most challenging grade to classify, reflecting its known morphological heterogeneity
- Automated WSI-level grade mapping could support pathologist workflows and enable more reproducible prostate cancer grading at scale

METHODS

- **Dataset:** PANDA dataset: 12,000 tissue patches from 10,616 biopsies (2,113 patients); patches classified into 4 categories: Gleason patterns 3, 4, 5, and non-cancerous tissue (normal epithelium, benign tissue, stroma)
- **Two-stage approach:** Stage 1 distinguishes cancerous vs. non-cancerous tissue; Stage 2 classifies individual Gleason patterns within cancerous regions
- **Model:** UNI FM fine-tuned on PANDA patches; predictions mapped back to WSI spatial coordinates with color-coded overlays
- **Evaluation:** 200 held-out test patches evaluated using accuracy, AUC, and Dice coefficient; stratified by Gleason pattern class



ACKNOWLEDGEMENTS

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